

QUESTION 2

QUICK PLAN - MINDMAP

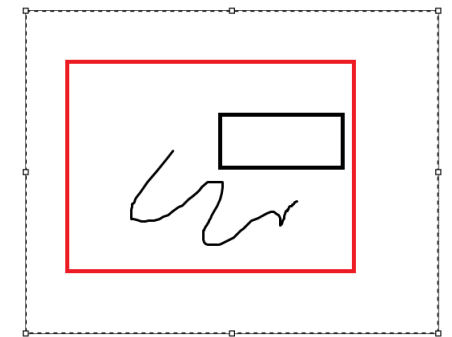
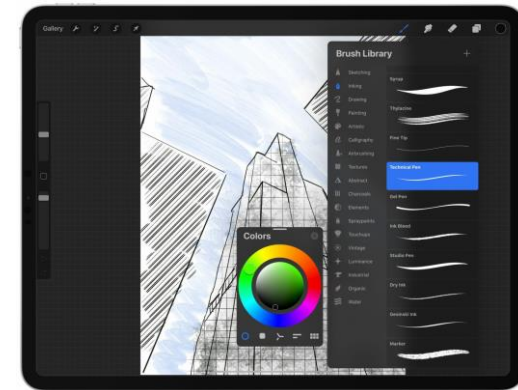
RESEARCH

PROCREATE

- FEATURES THAT STOOD OUT TO ME:
 - Layers
 - Text fonts, with glow, change size
 - An Isometric guide, Perspective Guide, Symmetry guide
 - 3D painting element
 - Blur
 - Different canvas sizes

MICROSOFT PAINT

- Research paint
- Research procreate
 - Editable shapes
 - Select shapes
 - Import Images
 - Zoom in zoom out



1. Add a line thickness slider in the same format of kaleidoscope tool which adjusts the strokeWeight so the thickness of the drawing tool can be adjusted. [createSlider() in the .options div]
2. Use unselecttool() function to make sure sliders don't appear when other tools are pressed

1. Create new JS file and add it to index.html <script>
2. Remove old unnecessary colour palette code and CSS and replace with colour picker CSS
3. Add the html colour picker by writing the type "color" (W3Schools)
4. createInput("rgb(0,0,0)", "color");
5. Keep base function the same

1. Add a line thickness slider in the same format of kaleidoscope tool which adjusts the strokeWeight so the thickness of the drawing tool can be adjusted [createSlider() in the .options div]
2. Use unselecttool() function to make sure sliders don't appear when other tools are pressed

1. Add eraser tool icon image to assets folder.
2. Create new JS file and add it to index.html using <script>
3. Add eraser to sidebar using this.icon
4. Use example provided by UoL for the line draw tool
5. Modify the setup() and the draw function to ensure it erases where mouse is pressed.
6. Change the colour to white so it erases it back to original colour of canvas stroke(255)
7. Ensure other tools don't change to white

LINE-DRAW
TOOL

COLOUR
PICKER TOOL

FREE-HAND
TOOL

ERASER
TOOL

DRAWING
APP

STAMP
TOOL

KALEIDOSCOPE
TOOL

SCISSORS TOOL

1. Add stamp tool icon image and stamp tool star image to assets folder.
2. Create new JS file and add it to index.html using <script>
3. Add stamp tool to sidebar
4. Create a number of stamp and size of stamp slider [createSlider()] in the .options div
5. Use example provided by UoL
6. Use unselecttool() function to make sure sliders don't appear when other tools are pressed
7. Modify the setup()
8. And the draw function to ensure images appear where the mouse is pressed
9. Adjust the stamp tool so that users can choose from a number of images by using an array. Each image is a number in the array. Modify populateOptions to contain a div for the stamp images. Add event listener to each image so when image is clicked it changes to that one on the canvas

KEY:

- : Completed
- : Started but more features need to be added
- : Not started at all

1. Add kaleidoscope tool icon image to assets folder.
2. Create new JS file and add it to index.html using <script>
3. Add kaleidoscope to sidebar using this.icon
4. Use example provided by youtube [p5.js Tutorial \(youtube.com\)](https://www.youtube.com/watch?v=p5jsTutorial)
5. Create a number of stamp and size of stamp slider [createSlider()] in the .options div
6. Use unselecttool() function to make sure sliders don't appear when other tools are pressed
7. Modify the setup() and the draw to allow lines to be drawn when mouse is pressed. Use pmouseX() and pmouseY().
8. Add a line thickness slider in the same format which adjusts the strokeWeight

Use populateOptions() to add the buttons

CLASS MAP

DRAWING APP

TOOL BOX

TOOL

All the following below inherits tool through icon: URL to the tool's icon imagename: Name of the tool

FREEHAND TOOL

PROPERTIES:

PreviousMouseX/Y:

Previous mouse locations

METHOD:

draw(): Draws a line from previous mouse location to the current mouse location using nested if statements
Import P5.sound library.

PlayDrawsound

LINETOOL

PROPERTIES:

- startMouseX: Starting X coordinate for the line
- startMouseY: Starting Y coordinate for the line
- METHODS:
- draw(): Draws straight lines between mouse press and release points using line(). Use curve() to create an option for curved line using populateOptions()

SPRAYCAN TOOL

PROPERTIES:

- points: Number of points in the spray
- spread: Spread radius of the spray
- METHODS:
- draw(): Draws a spray effect when mouse is pressed using for loop.

STAMP TOOL

PROPERTIES:

- Star: Image for the stamp
- starSizeSlider: Slider for star size
- nStarSlider: Slider for the number of stars
- METHODS:
- populateOptions(): Create options for stamp size and number of stars
- unselectTool(): Clears the options when another tool is selected
- draw(): Draws stars at the mouse position using if and for statements.

MIRROR DRAW TOOL

- PROPERTIES: lineOfSymmetry: Line of symmetry position
- previousMouseX/Y: Previous mouse X/Y coordinate
- previousOppositeMouseX/Y: Previous mirrored mouse X or Y coordinate
- METHODS: draw(): Draws mirrored lines based on the axis of symmetry
- calculateOpposite(n, a): Calculates the opposite coordinate of the mouse
- unselectTool(): Clears the options and the symmetry line
- populateOptions(): Populates options to create "make horizontal" and "make vertical" buttons that switch when either is selected using if statements

KALEIDOSCOPE TOOL

- PROPERTIES: colourSatSlider: Slider for colour saturation
- nReflection: Slider for the number of reflections
- lineThicknessSlider: Slider for line thickness
- METHODS: populateOptions(): creates options for colour saturation, number of reflections and line thickness
- unselectTool(): Clears options when another tool selected
- draw(): Draws kaleidoscope patterns based on slider values
- Translate(): moved centre point to the centre of canvas. For loop to draw plenty of lines with different angles

ERASER TOOL

- PROPERTIES:
- previousmouseX: Previous mouse X coordinate
- previousmouseY: Previous mouse Y coordinate
- METHODS:
- draw(): Draws white lines to erase the drawing when mouse is pressed using line().
- populateOptions(): creates options for eraser thickness
- unselectTool(): Clears options when another tool selected

SCISSORS TOOL

Not yet reached week 13 of studies yet in order to implement the code

PROPERTIES:

METHODS:

Draw()

Populateoptions()

unselectTool()

HELPER FUNCTIONS

- clearCanvas(): Clears the canvas
- saveImage(): Saves the canvas as an image

COLOUR PICKER

- PROPERTIES: selectedColour: Currently selected colour
- METHODS
- loadColours(): Loads the colour palette and sets up event listeners for colour changes

WIREFRAMES

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

STAMP SHAPE

Size of stamp

Number of stamp

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Make horizontal

Make vertical

Make Diagonal

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Colour

Thickness

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Colour

Thickness

CURVED LINE

STRAIGHT LINE

DASHED LINE

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Colour Saturation

Angle

Line Thickness

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Thickness

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

MY DRAWING APP

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

Kaleidoscope tool

Scissors tool

Colour

Thickness

MY DRAWING APP

Save

Clear

Layers

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

TEXT TOOL

Scissors tool

Colour

FONT 1

FONT 2

MY DRAWING APP

Save

Clear

Layers

Freehand tool

Line tool

Spray Can tool

Stamp tool

Mirror draw tool

Eraser tool

TEXT TOOL

Scissors tool

Layer 1

Layer 2

SYSTEM ARCHITECTURE

DRAWING APP

USER INTERFACE – What the user interacts with

HTML + CSS

Area where users draw by clicking the mouse

Tool Bar (Tool buttons on the side)

Colour Picker (RGB Colour Picker below)

Sliders for further adjustments

User Clicks

User selects tool

User selects colour

User adjusts slider

EVENT HANDLERS: example of a few events that will occur

handleMouseMove()

handleToolClick()

handleColourClick()

handleSliderClick()

User selects from
the tools below

Selecting a tool
will allow the
sliders/colours
to show as a
further option
or not

CORE LOGIC

Helper functions
Clear()
saveImage()

Setup- adds the tools functions to
the toolbox, creates canvas

Preload- loads images

Draw
Implements all the tools

TOOLBOX- manages all the tools and what will happen when tools selected and unselected

FreeHandTool

LineTool

EraserTool

SprayCanTool

KaleidoscopeTool

ScissorsTool

StampTool

MirrorDrawTool